

Supplementary Material

CliniCause Knowledge-Design Stage

ChatGPT Elicitation Summaries

Model metadata: ChatGPT 5.4 with extended reasoning

The same model setting was used for the MIMIC-III and PhysioNet sessions.

Standalone academic supplement

Contents

1	Purpose, scope, and handling of elicitation output	2
2	MIMIC-III session summary	2
3	PhysioNet session summary	3
4	Record-integrity notes	3

1 Purpose, scope, and handling of elicitation output

Both elicitation sessions used ChatGPT 5.4 with extended reasoning.

The CliniCause knowledge-design stage used ChatGPT to propose clinically and causally motivated design candidates for two ICU datasets. The model received dataset schema/column names and the mortality outcome, but no patient rows, empirical distributions, prevalence values, or causal effect estimates. It was asked to propose proxy variables, clinical cutoffs, temporal and aggregation rules, supporting literature, and dataset-specific directed acyclic graph (DAG) relationships.

The elicitation was an input to author review rather than an automated labeling or analysis step. The authors inspected the proposals and implemented the proposals they accepted. This supplement summarizes the records; it does not copy, quote, reproduce, correct, or complete any prompt or model response.

Session	Source record	Record scope summarized here
MIMIC-III	<code>thesis-writing/prompts-and-documents/mimic-prompt-running.pdf</code>	Latent-variable causal design using the MIMIC-III schema and mortality outcome.
PhysioNet	<code>thesis-writing/prompts-and-documents/physionet-prompt-running.pdf</code>	Latent-variable causal design using the PhysioNet ICU schema and in-hospital mortality outcome.

2 MIMIC-III session summary

Source-file provenance. `thesis-writing/prompts-and-documents/mimic-prompt-running.pdf`

Source PDF title. *mimic-prompt-running - Latent Causal Model MIMIC-III*

Source length and checksum. 44 pages; SHA-256 56271465a73f624151ba36b1b4c4a07ad0e4271a9bb7304256ede4155afbf79a.

The MIMIC-III session established a staged workflow: audit the available tables/variables; identify background, measurement, care-process, and outcome variables; select latent clinical states; operationalize those states with binary proxy rules; and then specify a sparse causal DAG. The response treated observed laboratory values, vital signs, and charted measures primarily as manifestations of latent physiology rather than as a dense network of direct causes.

The proposed design covered acute physiologic and organ-dysfunction domains, with attention to chronic baseline risk and global severity. It mapped observed measurements to candidate latent states, supplied implementable scoring or threshold-based proxy-label logic, and described missingness and measurement intensity as potentially informative care-process variables. It also included temporal safeguards against future information and outcome leakage, plus guidance on distinguishing confounders, mediators, and colliders for analyses of mortality.

The session further supplied candidate latent-to-latent, latent-to-observation, and latent-to-mortality relationships; pseudocode and an edge-list-oriented implementation representation; and literature-oriented rationale for clinical constructs and causal assumptions. The record flags limitations where a latent state is weakly measured or where care actions and documentation can complicate interpretation.

3 PhysioNet session summary

Source-file provenance. `thesis-writing/prompts-and-documents/physionet-prompt-running.pdf`

Source PDF title. *physionet-prompt-running - Causal Model for ICU Data*

Source length and checksum. 45 pages; SHA-256 `933c864a5c0679764d579d4617f5192c7f1db0a53cd74e5b0906b7297d98cb67`.

The PhysioNet session used the same latent-variable causal-design framework, but adapted the operationalization to the variables available in that ICU dataset. It audited demographic, vital-sign, laboratory, and care-related fields; proposed candidate latent physiologic and organ-dysfunction states; and defined binary proxy-label rules with clinically motivated thresholds and aggregation over observed signals.

The response explicitly emphasized the dataset’s information limits: several clinical concepts cannot be directly identified when diagnoses, medications, procedures, microbiology, comorbidity coding, admission diagnosis, or clinician intent are absent. Accordingly, it presented states such as sepsis, shock, and cardiac injury as proxy constructs rather than directly observed ground truth. It also considered missingness and measurement intensity, temporal leakage, and the role of care processes when interpreting measurements.

The proposed DAG separated background/case-mix factors, latent acute physiology, observed measurements, care or measurement processes, and in-hospital mortality. It included causal-analysis guidance on adjustment, mediators, and collider risk; code-oriented representations for implementation; and supporting clinical and causal literature directions. The session marked the limits of available evidence and treated proposed label rules as operational approximations for subsequent author review.

4 Record-integrity notes

The two source records are distinct PDFs with distinct titles, session links, page counts, and checksums. Their requested staged methodology is closely parallel, while the dataset audits and later operational details are dataset-specific. No duplicate source file was used in preparing this summary.

Text extraction identifies a trailing blank page in each source record (page 44 for MIMIC-III and page 45 for PhysioNet); this appears to be a source-export artifact, not missing prompt material. The source records themselves do not provide patient-level data, prevalence summaries, distributions, or causal effect estimates. Model-setting metadata is stated in this supplement as provided for the study: ChatGPT 5.4 with extended reasoning.